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Analysis of complex dielectric spectra. I. One-dimensional derivative techniques and three-dimensional modelling

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Abstract

Several strategies for the analysis of isochronal and isothermal spectra as well as complete three-dimensional (3D) spectra are outlined and applied to data of (glass forming) liquids and polymers. Conduction-free loss spectra are calculated using a compact solution of the Kramers–Kronig transformation and an approximation based on $\partial\epsilon'/\partial \ln \omega$. The usefulness of the dielectric modulus for data analysis is also evaluated. Apart from $\partial\epsilon'/\partial \ln \omega$, other useful derivatives in the frequency and temperature domain are treated as well. Finally, we describe the evaluation of 3D-relaxation maps by means of 3D fitting. © 2002 Elsevier Science B.V. All rights reserved.

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1. Introduction

Owing to the progress in electronic instrumentation, dielectric relaxation spectroscopy (DRS) has become a popular and powerful technique for studying the relaxation dynamics of almost any kind of material. Modern measurement systems allow the acquisition of relaxation spectra over a wide range in frequency and temperature with a high accuracy. The availability of fully automated spectrometers has two important consequences:

- (i) huge amounts of data have to be managed. This requires an effective analysis.
- (ii) the enhanced accuracy and data density enable reliable modeling with a sum of relaxation functions as is necessary for evaluating complex (e.g. multiphase or biological) materials.

In this paper we will present tools for the analysis of 2D spectra as well as complete 3D-dielectric spectra, whereby both the permittivity and its reciprocal—the dielectric modulus—are considered. Apart from common fit techniques for isothermal spectra using a double stretched relaxation function (e.g. Havriliak–Negami, HN), we will discuss a convenient numerical version of the Kramers–Kronig transform [1,2] as well as the useful derivatives $\partial\epsilon'/\partial \ln \omega$ and $\partial\epsilon'/\partial T$ [3–5]. These 1D derivatives also form the basis for a

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bivariate differential sampling technique, which is described in a separate contribution [6].

Finally, the analysis of three-dimensional (3D) dielectric ‘landscapes’, e.g. $\varepsilon''(\omega, T)$ by means of generalized HN functions is addressed [7,8].

The capabilities and limitations of all approaches will be illustrated for glass forming liquids and for polymers with multiple relaxations.

2. Representations of dielectric spectra

Although dielectric spectroscopy yields at least two dielectric spectra viz. for ε' and for ε'' , the analysis and discussion of the relaxation properties is mainly based on the loss spectra ε'' . Two reasons for this practice are

- (i) The dispersion curves often reveal more details and therefore allow a better evaluation of complex data (e.g. overlapping peaks).
- (ii) Since ε' and ε'' are interrelated by the Kramers–Kronig transforms, the two quantities ε' and ε'' are equivalent with respect to their information.

The KK integral transforms for the real and imaginary part of dielectric relaxations read [9–13]

$$\varepsilon'(\omega_0) = \varepsilon_\infty + \frac{2}{\pi} \int_0^\infty \varepsilon''(\omega) \frac{\omega}{\omega^2 - \omega_0^2} d\omega, \quad (1)$$

$$\varepsilon''(\omega_0) = \frac{\sigma_{dc}}{\varepsilon_v \omega_0} + \frac{2}{\pi} \int_0^\infty \varepsilon'(\omega) \frac{\omega_0}{\omega^2 - \omega_0^2} d\omega. \quad (2)$$

The extra loss term in Eq. (2) is due to (ionic) Ohmic conduction. Ion conduction often shows up above the glass-transition temperature T_g and might obscure loss peaks of dipolar origin, in particular at low frequencies. The Ohmic conduction should therefore be eliminated in order to elucidate low-frequency relaxation processes.

2.1. Compact numerical version of the KK transform

An elegant way to remove Ohmic (i.e. frequency independent) conduction from measured loss spectra uses Kramers–Kronig relation Eq. (2), whereby we transform the real part ε' into an imaginary

part ε''_{KK} that is solely based on relaxation phenomena. By virtue of this, ε''_{KK} lacks the Ohmic conduction term. It can be calculated by numerical techniques described in [1–3].

A quite accurate approximation is

$$\varepsilon''_{KK}(\omega) = \sum_{k=1}^4 a_k [\varepsilon'(\omega/2^k) - \varepsilon'(2^k/\omega)], \quad (3)$$

where $a_k = 0.4453, 0.22726, -0.11, 0.13458$. Special coefficients are needed to find ε''_{KK} at the beginning and end of the data-array. Alternatively, ε''_{KK} can be calculated by inverting a tridiagonal set of ε' -values; for details see [1,2].

Fig. 1 illustrates the agreement between the measured loss ε'' (○) and ε''_{KK} (●) for the α -relaxation peak of the low molecular glass former *m*-toluidine (3-amino toluene, $H_2NC_6H_4CH_3$). At low frequencies the two curves start to deviate as can be anticipated in view of the strong conduction losses in ε'' -measured.

2.2. First order approximation of the KK transform

An alternative to the numerical Kramers–Kronig transform is based on the logarithmic derivative

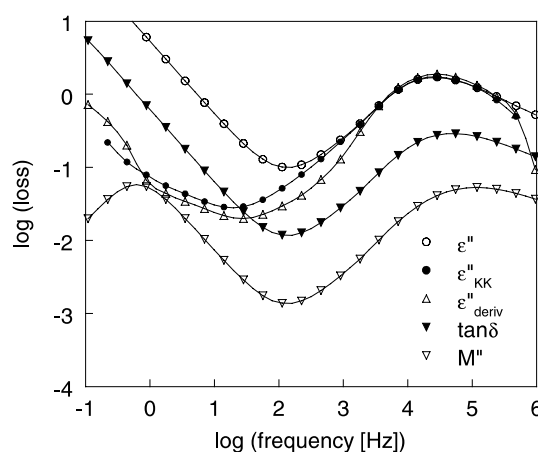


Fig. 1. The α -relaxation peak of *m*-toluidine (3-amino toluene, $H_2NC_6H_4CH_3$) at -65°C represented in five ways: (a) measured loss ε'' , (b) ε''_{KK} computed from ε' , (c) $\varepsilon''_{\text{deriv}}$ also computed from ε' (d) measured $\tan \delta$ and (e) M'' calculated from ε' and ε'' .

$$\varepsilon''_{\text{der}} = -\frac{\pi}{2} \frac{\partial \varepsilon'(\omega)}{\partial \ln \omega} \approx \varepsilon''_{\text{rel}} \quad (4)$$

which approximately equals the Ohmic-conduction-free dielectric loss $\varepsilon''_{\text{rel}}$ for rather broad peaks, like those of the α -transition or the secondary relaxations [5]. For non-distributed, Debye relaxations, i.e. for single-relaxation time processes, the derivative results in peaks that are sharper:

$$\frac{\partial \varepsilon'(\omega)}{\partial \ln \omega} = -\frac{\varepsilon''^2}{\varepsilon''_{\text{max}}} \quad (5)$$

For the double stretched, HN function [14] we have for ε'' and $\partial \varepsilon' / \partial \ln \omega$

$$\begin{aligned} \varepsilon'' &= -\text{Im} \left\{ \frac{\Delta \varepsilon}{(1 + (i\omega\tau)^a)^b} \right\} \\ &= \Delta \varepsilon \left[1 + 2(\omega\tau)^a \cos(\pi a/2) + (\omega\tau)^{2a} \right]^{-b/2} \\ &\quad \sin(b\theta_{\text{HN}}) \end{aligned} \quad (6)$$

with $\Delta \varepsilon = \varepsilon_s - \varepsilon_\infty$ and

$$\theta_{\text{HN}} = \arctan [\sin(\pi a/2) / ((\omega\tau)^{-a} + \cos(\pi a/2))],$$

while

$$\frac{\partial \varepsilon'_{\text{HN}}}{\partial \ln \omega} = -\frac{ab\Delta \varepsilon (\omega\tau)^a \cos[a\pi/2 - (1+b)\theta_{\text{HN}}]}{\left[1 + 2(\omega\tau)^a \cos(\pi a/2) + (\omega\tau)^{2a} \right]^{(1+b)/2}} \quad (7)$$

The shape parameters a and b are usually constrained to $0 < a, b \leq 1$, however b might exceed 1, provided that $ab \leq 1$.

The ratio of the maxima of $\varepsilon''_{\text{der}}$ and ε'' is given by

$$\frac{\varepsilon''_{\text{der,max}}}{\varepsilon''_{\text{max}}} = \frac{\pi a/2}{\sin(\pi a/2)} \quad (8)$$

if $b = 1$ (Cole–Cole). The peak sharpening is demonstrated in Fig. 2. Hence by using $\partial \varepsilon' / \partial \ln \omega$, nearby peaks can be better resolved.

Although there are several options for implementing Eq. (4) to differentiate the experimental data, one should realize that derivatives are sensitive to inaccuracies, and noise. Therefore, we have chosen a numerical technique based on a low pass quadratic least squares filter [15,16]. We use ε' -data equally spaced on the $\log \omega$ scale

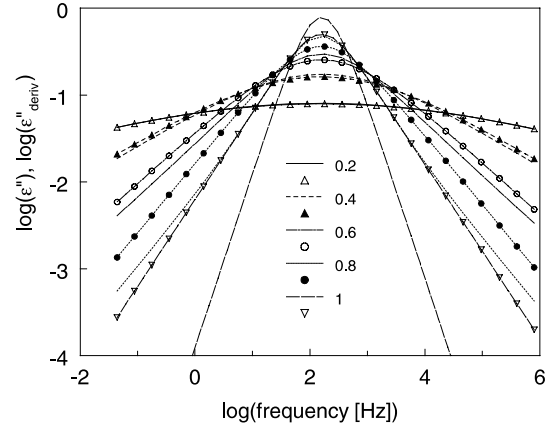


Fig. 2. Comparison of dielectric loss ε'' (lines with symbols) and $\varepsilon''_{\text{der}}$ (lines) for a symmetric ($b = 1$) HN function (Eq. (6)) for various shape parameters a (0.2, 0.4, 0.6, 0.8, 1). We took $\Delta \varepsilon = 1$ and $\tau = 0.001$ s.

$$\varepsilon''_{\text{der}}(\omega) = \sum_{k=-2}^2 b_k \varepsilon'(r^k \omega) \quad (9)$$

where $b_k = (-2, -1, 0, 1, 2)/10 \ln r$, with a logarithmic spacing $r = \sqrt{2}$, $10^{0.15}$ or 2.

We have used logarithmic-equidistant five point splines as well [17]. In that case

$$\varepsilon''_{\text{der}}(\omega) = \sum_{k=-2}^2 c_k \varepsilon'(r^k \omega), \quad (10)$$

where $c_k = (1, -6, 0, 6, -1)/8 \ln r$. At the start and the end of the ε' -array we use special coefficients, so that all data can be converted.

2.3. Dielectric modulus

Recently, there is a revival in the use of the reciprocal permittivity or dielectric modulus [10]. Although this quantity can be measured directly [18], it can also be readily found from

$$M^*(\omega) = 1/\varepsilon^*(\omega) = 1/(\varepsilon' - i\varepsilon'') \quad (11)$$

and so

$$\begin{aligned} M^*(\omega) &= M'(\omega) + iM''(\omega) \\ &= \frac{\varepsilon'}{\varepsilon'^2 + \varepsilon''^2} + i \frac{\varepsilon''}{\varepsilon'^2 + \varepsilon''^2}. \end{aligned} \quad (12)$$

Note that $\tan \delta$ are the same for M^* and ε^* , since

$$\frac{\varepsilon''}{\varepsilon'} = \frac{M''}{M'} = \tan \delta. \quad (13)$$

The imaginary part M'' displays the interesting property that it ‘converts’ the low-frequency steady increase in ε'' caused by ionic conduction, into a specific ‘conduction’ peak, cf. Fig. 1, the maximum of which yields the Ohmic relaxation time $\tau_\sigma = \varepsilon_\infty \varepsilon_v / \sigma$. The α -relaxation peak in ε'' also shows up as a peak in M'' in Fig. 1, but it is shifted to higher frequencies.

Analytical modeling of M^* is less common than that for ε^* . However, we can readily derive from Eq. (11) for a Debye relaxation or single-relaxation time response

$$\frac{M_\infty - M^*(\omega)}{M_\infty - M_s} = \frac{1}{1 + \frac{\varepsilon_\infty}{\varepsilon_s} i\omega\tau}, \quad (14)$$

where $M_\infty = 1/\varepsilon_\infty$ and $M_s = 1/\varepsilon_s$. Eq. (14) explains the shift of the α -peak to higher frequencies, because the M'' -peak will emerge at

$$(\omega\tau)_{M''_{\max}} = \varepsilon_s/\varepsilon_\infty \quad (15)$$

rather than at $(\omega\tau)_{\max} = 1$. For a Debye relaxation this shift will on a log-scale be twice that of $\tan \delta$, since $(\omega\tau)_{\tan \delta, \max} = (\varepsilon_s/\varepsilon_\infty)^{1/2}$.

We can also easily account for Ohmic conduction. This modifies Eq. (14) to

$$M^*(\omega) = \frac{1}{\varepsilon_\infty + \frac{\varepsilon_s - \varepsilon_\infty}{1 + i\omega\tau} - \frac{i\sigma}{\varepsilon_v \omega}}. \quad (16)$$

At low frequencies and for high temperatures the dipolar relaxation time becomes very small. We are then left with the conduction term and can approximate Eq. (16) by

$$M^*(\omega) \simeq \frac{1}{\varepsilon_s - i\varepsilon_\infty/\omega\tau_c}. \quad (17)$$

According to Eq. (17) the conduction peak in M'' is reached at

$$(\omega\tau_c)_{M''_{\max}} = \varepsilon_\infty/\varepsilon_s \quad (18)$$

which coincides with $\tan \delta = \varepsilon''/\varepsilon' = 1$ if the conduction prevails. The differences in position of $\varepsilon''_{\max}$, etc. for the α -relaxation peak can be seen in Fig. 1.

The analysis of M^* can be extended to other relaxation functions, such as that of HN, for which we obtain

$$\frac{M_\infty - M^*(\omega)}{M_\infty - M_s} = \frac{1}{1 + \frac{\varepsilon_\infty}{\varepsilon_s} \left[(1 + (i\omega\tau)^a)^b - 1 \right]}. \quad (19)$$

Eq. (19) holds for the relaxation part of the M^* -response (i.e. the α -peak in Fig. 1), because we neglected the conduction. If we combine a HN relaxation with conduction we get

$$M^*(\omega) = \frac{1}{\varepsilon_\infty + \frac{\varepsilon_s - \varepsilon_\infty}{[1 + (i\omega\tau)^a]^b} - \frac{i\sigma}{\varepsilon_v \omega}}. \quad (20)$$

This equation again reduces to Eq. (17) if the conduction becomes dominant.

By putting $b = 1$ into Eq. (19) we find M^* for Cole–Cole

$$\frac{M_\infty - M^*(\omega)}{M_\infty - M_s} = \frac{1}{1 + \frac{\varepsilon_\infty}{\varepsilon_s} (i\omega\tau)^a}. \quad (21)$$

Eq. (21) is very similar to the expression of ε^* for Cole–Cole, compare Eq. (6) with $b = 1$, merely the location of the two loss peaks will be different. The M'' -peak is found at

$$(\omega\tau)_{M''_{\max}}^a = \varepsilon_s/\varepsilon_\infty, \quad (22)$$

while the ε'' and $\tan \delta$ peak are centered at $(\omega\tau)_{\varepsilon''_{\max}} = 1$ and $(\omega\tau)_{\tan \delta, \max}^{2a} = \varepsilon_s/\varepsilon_\infty$ respectively.

Returning to a HN-relaxation function, we can expect to find its M'' -peak to show up at about

$$(\omega\tau)_{M''_{\max}}^{ab} \simeq \varepsilon_s/\varepsilon_\infty, \quad (23)$$

whereas the ε'' -peak for HN is located at [7,19]

$$(\omega\tau)_{\varepsilon''_{\max}}^a = \frac{\sin(\pi a/(2+2b))}{\sin(\pi ab/(2+2b))}. \quad (24)$$

Since $a, ab < 1$ the frequency shift of the M'' -peak will be substantial for a distributed relaxation. So a M'' -based analysis can be expected to give a better separation between dipole relaxations and conduction dominated losses than one based on ε'' . This is substantiated by the curves shown in Fig. 1.

Fig. 3 compares the relaxation time for ε'' and the M'' -peaks of the α -process of *m*-toluidine at different temperatures. As expected both show the same T -dependence, apart from the above calculated shift. The same holds for the temperature

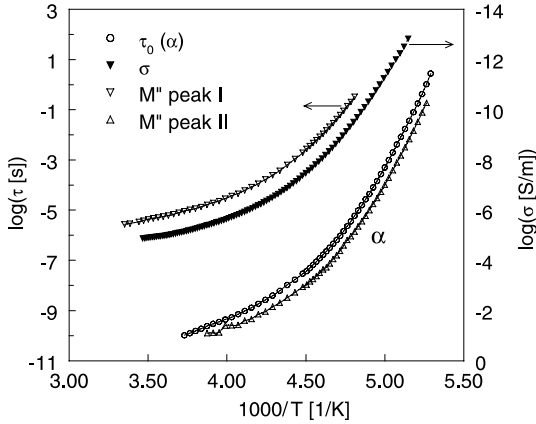


Fig. 3. Arrhenius plot for *m*-toluidine showing the log of the α -peak relaxation time τ_m obtained from the measured ε'' -peaks and from the calculated M'' -relaxation peaks vs. $1/T$. The figure also depicts the course of the conductivity σ (full triangles) and of the Ohmic relaxation time τ_σ obtained from the low-frequency M'' -conduction peak.

dependence of the conductivity σ and the Ohmic relaxation time τ_σ .

3. Derivatives in the frequency domain

3.1. Suppression of effect of electrode polarization

Electrode polarization (EP) often is an undesirable effect in DRS which hampers the proper analysis of slow dipolar relaxation processes in moderately to highly conducting materials, such as polymer melts. The reason for EP is the (partial) blocking of the charge exchange at the sample/electrode interfaces, which results in the formation of two double layers. These double layers give rise to large capacitances in series to the conducting bulk of the sample. This manifests itself in a high apparent dielectric constant typically in the range of 10^2 – 10^6 .

A theoretical analysis of EP has been given by several authors [11,12,20,21]. Theories about the permittivity of ionic solids and suspensions are also useful [21–23]. Like others, Coelho [20], who considered both diffusion and drift of ions in a sample with blocking electrodes, has shown that

EP can be approximated quite closely by a Debye model

$$\varepsilon_{EP}^*(\omega, T) \simeq \varepsilon_\infty + \frac{\varepsilon_\infty \delta(T)}{1 + i\delta(T)\omega\tau_c(T)}, \quad (25)$$

where $\delta = L/2L_D(T)$, with L the thickness of the sample and L_D the Debye length. Eq. (25) is a good approximation, for $\delta \geq 10$. The full expression is given in [20], it reads

$$\varepsilon_{EP}^*(\omega, T) = \varepsilon_\infty + \frac{\varepsilon_s - \varepsilon_\infty}{1 + i\omega\tau'(T)}, \quad (26)$$

where $\varepsilon_s/\varepsilon_\infty = \delta \coth \delta$ and $\tau' = 0.5[3\delta \coth \delta - \delta^2/(\delta \coth \delta - 1)]\tau_c$. Coelho's Debye, single-relaxation time expression Eq. (26) holds for small and large δ 's. For $\delta \geq 3$ his ε_s and τ' can be approximated more simply by $\varepsilon_s/\varepsilon_\infty \simeq \delta$ and $\tau' \simeq (1.00465\delta - 0.6423)\tau_c$.

Since EP involves charging/discharging of the two double layer capacitances, the thickness of which is given by L_D , it is not surprising that both its strength and effective relaxation time scale approximately, linearly with δ , i.e. with $L/2$ over L_D . In line with the Debye model of Eq. (25) we have

$$\Delta\varepsilon_{EP} = \varepsilon_s - \varepsilon_\infty \simeq \varepsilon_\infty \delta = \varepsilon_\infty L/2L_D \quad (27)$$

and

$$\tau_{EP} \simeq \delta\tau_c = \frac{\varepsilon_\infty \varepsilon_v}{\sigma} \frac{L}{2L_D}. \quad (28)$$

These simple approximations enable one in return to determine the Debye length of a system. This knowledge can be exploited to determine the size of inclusions or fillers in heterogeneous materials [24,25].

Regarding ε_{EP}'' the $L/2L_D$ scaling results in a high frequency slope of the EP loss peak of -1 , which is independent of the sample thickness L . This is illustrated in Fig. 4 (top).

Interestingly, the curves of ε_{der}'' depicted below in Fig. 4 do not join on the high frequency side since they fall off with $\propto \omega^{-2}$. This results in an apparent shift of the interfering EP peak towards lower frequencies upon increase of L [23]. Consequently, the use of the logarithmic derivative enables one to unravel genuine relaxation processes by optimizing the sample thickness.

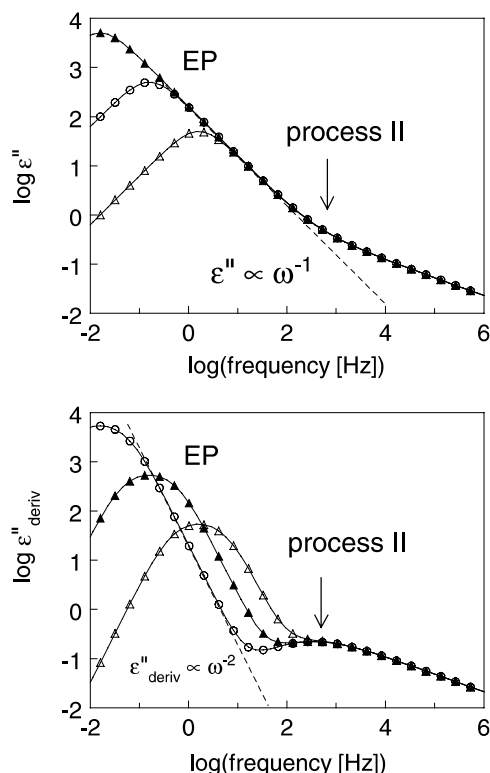


Fig. 4. Two model spectra simulating EP for three samples thicknesses (L , $10L$, $100L$) in combination with a high-frequency dipole peak (II). The EP peak was calculated with a Debye model, taking $\Delta\epsilon_{EP} = 10^2$, 10^3 and 10^4 . For the dipole relaxation (II) a HN function was used with $\tau = 10^{-3}$ s, $\Delta\epsilon = 1$, $a = 0.6$ and $b = 0.6$. The ϵ''_{der} spectra are clearly sharper.

3.2. Deconvolution of complex relaxation spectra

Apart from being beneficial for handling a strong EP, the logarithmic derivative also turns out to be useful for polymers which exhibit low-frequency relaxations (that are slower than the main or α -relaxation) alongside an appreciable Ohmic conductivity. This is often the case for side-chain liquid crystalline polymers [4], an example is presented in Fig. 5.

In the loss spectrum (top), we can identify two relaxation processes above T_g : the α -process related to the dynamic glass transition of the polyurethane backbone and a slower relaxation λ_1 involving the rigid, mesogenic nitrostilbene group.

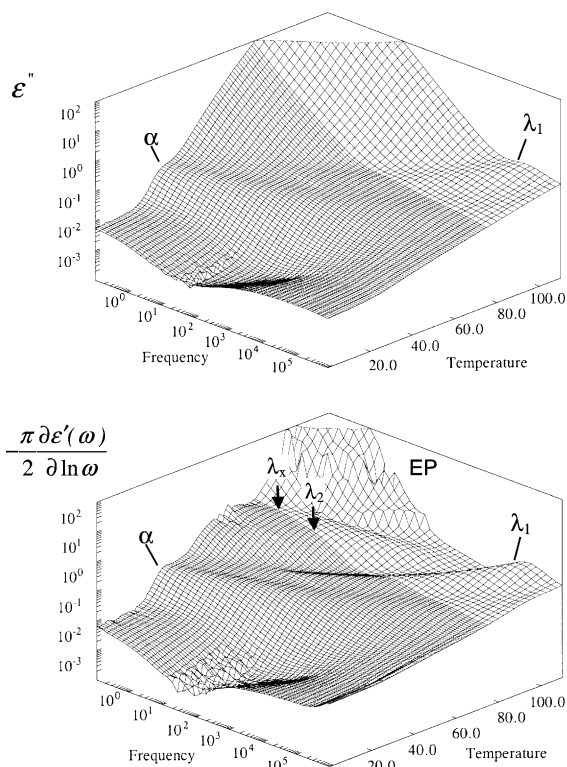


Fig. 5. Loss ϵ'' (top) and ϵ''_{der} (bottom) of a side-chain liquid crystalline polyurethane [27] measured during heating. The derivative reveals two additional slow mesogenic relaxation processes λ_2 and λ_x (black arrows), which could be sufficiently set apart from the EP loss by using an extra thick sample ($L = 500 \mu\text{m}$).

Interestingly, the ϵ''_{der} spectra reveal two additional relaxation processes of the mesogenic group on the low-frequency side of the λ_1 process,¹ which are missing in the ϵ'' -curves. A further comparison between both quantities is made in Fig. 6, which shows a clear peak sharpening around the λ_1 -relaxation, indicating that this relaxation behaves Debye-like.

Figs. 1 and 6 further show the restrictions of the derivative approach: (i) slight inaccuracies of the numerical calculations at the very edges of the frequency range, (ii) deviations at high frequencies

¹ We label peaks due to mesogenic groups with λ , whereby λ refers to the liquid crystalline behaviour. In the literature, these mesogenic relaxations are often called α and δ .

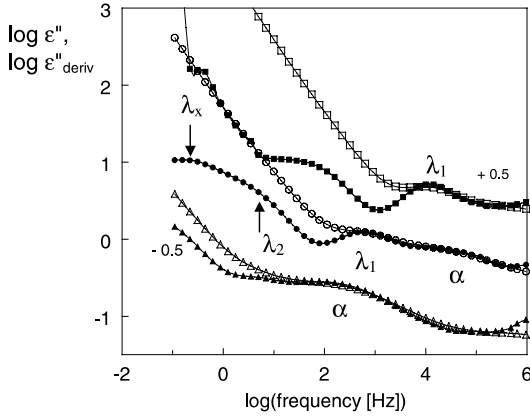


Fig. 6. Dielectric loss ε'' (open symbols) and $\varepsilon''_{\text{deriv}}$ (filled symbols) spectra at three temperatures selected from Fig. 5. For clarity, the top and bottom curves are shifted vertically by $\log \varepsilon'' = \pm 0.5$. The arrows indicate the two extra relaxations of the nitrostilbene group uncovered by $\varepsilon''_{\text{deriv}}$.

due to instrumental phase inaccuracies as well as more noise in $\varepsilon''_{\text{deriv}}$ at low frequencies caused by errors in the ε' -data.

3.3. Other useful derivatives in the frequency domain

In addition to $\partial \varepsilon' / \partial \ln \omega$, we have applied another derivative $\partial \ln \varepsilon'' / \partial \ln \omega$ which yields the local slope of loss spectra in a double-logarithmic representation. Fig. 7 shows the temperature dependence of ε'' of propylene glycol (PG) at different frequencies along with the local slope $-\partial \ln \varepsilon'' / \partial \ln \omega$ determined from isothermal spectra. As is typical for a glass forming liquid, PG shows loss curves featuring two prominent facts, a strong ionic conduction at high temperatures and a well resolved α -relaxation, which are usually modeled by a sum of a power law and an HN function

$$\varepsilon'' = \frac{\sigma}{\varepsilon_0 \omega^s} + \text{Im} \left\{ \frac{\Delta \varepsilon}{(1 + (i\omega\tau)^a)^b} \right\}. \quad (29)$$

We see from Fig. 7 that $\partial \ln \varepsilon'' / \partial \ln \omega$ directly produces the ‘shape parameters’, namely the power of the conduction term (usually $s \approx 1$) and the shape of the α -process determined by a and $-ab$. In this case, we find $s = 1$, $a \approx 0.95$ and $b \approx 0.68$ – 0.75 . Note that a and b are not constant.

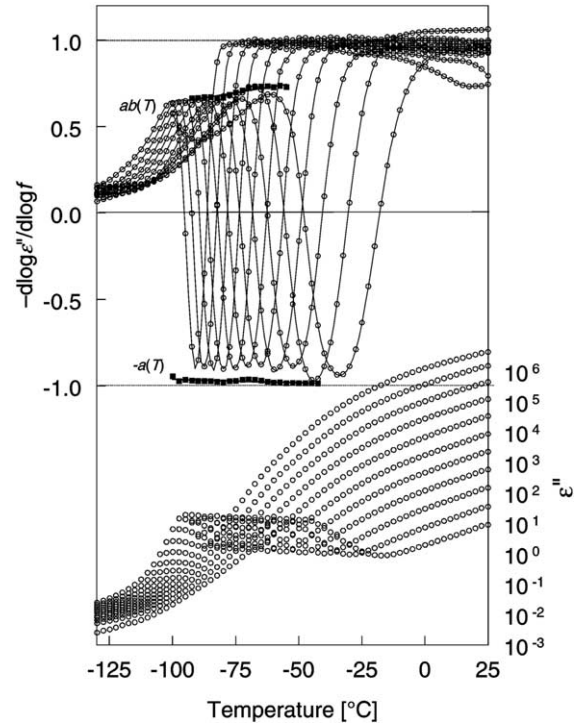


Fig. 7. Dielectric loss ε'' (bottom) and double derivative $-\partial \log \varepsilon'' / \partial \log \omega$ (top) of PG (1,2-propane diol, $\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{OH}$) at various frequencies. The shape-parameters a and b are found to increase a little with T , as does $\varepsilon''_{\text{max}}$ in the lower curves. The black squares correspond to the a - and b -values derived from the HN fits of the ε'' -peaks, the agreement is quite satisfactory.

The position of the loss maxima, at which Eq. (24) is satisfied, can also easily be found, viz. from the condition $\partial \ln \varepsilon'' / \partial \ln \omega = 0$. Consequently, a plot of $\partial \ln \varepsilon'' / \partial \ln \omega$ gives direct access to the shape parameters of relaxation processes and it allows to distinguish ‘Ohmic’ from ‘non-Ohmic’, frequency dependent conduction without time-consuming fit procedures.

4. Three-dimensional modeling of dielectric spectra

Dielectric relaxations at sub- T_g temperatures usually originate from molecular motions that are restricted to the scale of few bond lengths. The activation energy of these processes is 10–50 kJ/mol depending on the bond rotational potentials.

Their relaxation times are almost always distributed due to a variation in molecular environments (local disorder). Consequently, secondary processes manifest themselves as broad relaxation peaks, which can often be modeled with a HN function [14] (see Eq. (6)).

Dielectric spectra containing several relaxation processes ($\gamma, \beta, \alpha \dots$) can be represented as a sum of HN functions, each characterized by a relaxation strength $\Delta\epsilon$, a mean relaxation time τ and two shape parameters a and b . Since broad peaks might extend over several frequency decades, multiple relaxation processes are in practice difficult to separate in the frequency domain.

Dielectric loss curves obtained by temperature scans i.e. as a function of temperature often display a more detailed structure. In order to utilize this advantage, we have fitted the loss curve ϵ'' with a HN function in the T -domain

$$\epsilon''(\omega_0, T) = \text{Im} \left\{ \frac{\Delta\epsilon(T)}{[1 + (i\omega_0\tau(T))^{a(T)}]^{b(T)}} \right\}. \quad (30)$$

The analytical expression for ϵ'' and the position of its maximum have been given in Eqs. (6) and (24). One can readily verify with the latter that ϵ''_{max} might be reached well above $\omega\tau = 1$, for asymmetric HN functions ($b \neq 1$). We further derived for the height of the loss maximum

$$\epsilon''_{\text{max, hn}} = \Delta\epsilon \frac{\sin(\pi ab/(2 + 2b))^{1+b}}{\sin(\pi a/2)^b}. \quad (31)$$

In addition to the T -dependent mean relaxation time $\tau(T)$, we have introduced in Eq. (30) a temperature-dependent relaxation strength $\Delta\epsilon$ and T -dependent shape parameters a and b , in agreement with the experimental results of Fig. 7. The common assumption for $\tau(T)$ is an Arrhenius behavior: $\tau(T) = \tau_\infty \exp(E_a/kT)$ or a VFT dependence (cf. Eq. (33)) [26].

For $\Delta\epsilon$ we have used a linear relation with respect to the inverse temperature:

$$\Delta\epsilon(T) = \Delta\epsilon_\infty(1 + m_e/T). \quad (32)$$

Eq. (32) can account for a relaxation strength that increases ($m_e < 0$, typical for secondary relaxations) or decreases ($m_e > 0$, α -process and fully

relaxed behavior) with temperature. The quantity $\Delta\epsilon_\infty$ corresponds to the limit of $\Delta\epsilon$ for $T \rightarrow \infty$. However, any reference temperature can be used, e.g. the peak maximum $T_{\text{ref}} = T_{\text{max}}$. This leads to

$$\Delta\epsilon(T) = \Delta\epsilon(T_{\text{ref}})(1 + m_e^*(1/T - 1/T_{\text{ref}})). \quad (33)$$

We have chosen similar expressions for the shape parameter a :

$$a(T) = a_\infty(1 - m_a/T) \quad (34)$$

or

$$a(T) = a(T_{\text{ref}})(1 - m_a^*(1/T - 1/T_{\text{ref}})). \quad (35)$$

Eqs. (34) and (35) are particularly useful for describing the peak shape of secondary relaxations in polymers, which often become narrower towards higher temperatures as a consequence of an underlying distribution in activation energies. Note that a plain HN function rather relies on a distribution in prefactors. The adaptations made partially correct for this restriction.

Although Eq. (30) can model a loss curve ϵ'' very well, one should be aware that the shape and position of a peak in the temperature domain depends not only on the shape parameters a, b but on the activation parameters (τ_∞ and E_a) as well. This might yield less reliable fit results, if no additional

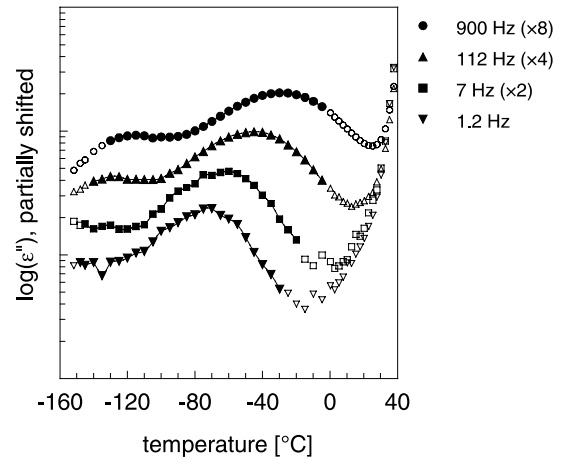


Fig. 8. Loss curves ϵ'' at four frequencies of a LC-polycarbonate [4]. The data points that were used in the 3D-fit procedure are highlighted by full symbols. The curves have been shifted vertically for more clarity.

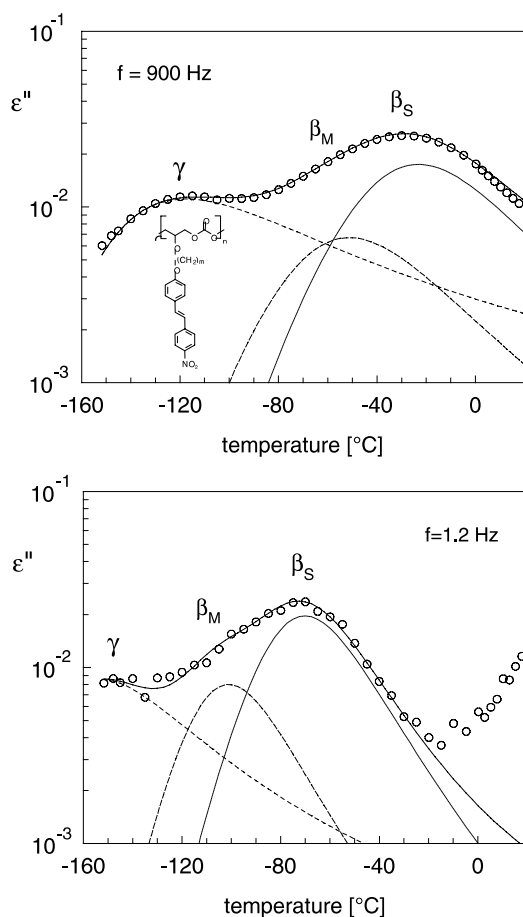


Fig. 9. Measured and fitted loss curves $\varepsilon''(T)$ at $f = 1.2$ (below) and 900 Hz (top) of a LC-polycarbonate which was fitted using Arrhenius kinetics with the same set of three temperature-dependent HN functions. The fitting unveiled an intermediate peak β_M that is almost invisible in the original data.

information like the real part ε' is considered in the fit program.

Therefore we apply a non-linear least squares fit based on the Levenberg–Marquardt method [7,8,12], which uses loss data taken at different frequencies and temperatures in a ‘global’ 3D-fit procedure and thus minimizes the expression:

$$\sum_T \sum_\omega \left(\varepsilon''_{\text{exp}} - \varepsilon'' \right)^2 \rightarrow \text{Min.} \quad (36)$$

Here the quantity ε'' represents a sum of HN functions. We combine e.g. simultaneously 4–6 temperature-scans containing 25–28 temperature

points each, at 4–6 frequencies differing by say a decade (cf. Fig. 8). The conduction-free loss $\varepsilon''_{\text{der}}$ can be fitted in a similar manner.

An example of the goodness of this 3D fitting is given in Fig. 9, which shows the measured and fitted loss curves for a liquid crystalline polycarbonate with a $(\text{CH}_2)_6$ spacer [4] at the 2 outmost frequencies. The fitted curves are represented by the same set of three symmetric HN functions ($b = 1$) and with the same set of Arrhenius parameters. In this particular case, the ‘global’ fit technique dug up a hidden weak (β_M) relaxation process with an accuracy of about 10% in the activation parameters E_a and τ_∞ .

5. Conclusions

We have illustrated the use of derivatives of dielectric quantities like ε' in order to achieve a more complete analysis of the kinetics of multiple relaxation processes and Ohmic conduction losses.

Other advantages of the use of derivatives as research tool in DRS is that the resolution of nearby loss peaks can be improved, that the losses by Ohmic conduction can be eliminated and that the effect of EP can be suppressed. The better separation of dipolar relaxations and of conduction losses in the dielectric modulus spectra have also been analyzed.

We have further outlined the use of 3D modelling of dielectric spectra in order to find more trustworthy parameters. We have thereby generalized the (HN) fit function by allowing its parameters to be temperature dependent.

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